

Estimating Housing Demand for the Market-for-Regulation Project

Layer I: A Random-Coefficients Sorting Model with Tenure Choice — Data, the Soil Cost-Shifter Instrument, and Judgment Calls

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Abstract

This memo specifies the demand layer in full. It resolves the three questions left open in the operationalization note — the *unit of demand*, the *instrument*, and *tenure choice* — and lays out the data, the estimator, and the judgment calls. The chosen architecture: a Berry–Levinsohn–Pakes random-coefficients discrete-choice model (Berry et al., 1995, Berry, 1994) estimated with modern best practices (Conlon and Gortmaker, 2020), in its residential-sorting form (Bayer et al., 2007), where households jointly choose a *sub-jurisdictional location* and a *tenure mode* (own / rent). Two recent institutional-investor job-market papers — Coven [2025] and Barbieri and Dobbels [2026] — are used throughout as concrete templates, because each estimates exactly this kind of tenure-differentiated BLP housing demand and each is close to your data environment (CoreLogic/Verisk property data; ACS; a builder/supply response). The novel contribution is an instrument: the foundation-cost component of soil (expansive clay / bearing capacity / shrink-swell), residualized against the agglomeration and amenity channels that the existing geology-IV literature (Rosenthal and Strange, 2008) actually exploits.

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1 The three resolved questions

1. Full BLP — committed. The demand side is a full random-coefficients logit, not a plain or nested logit. This is treated as settled and the memo builds around it. The justification is both substantive (realistic substitution patterns are needed for credible counterfactuals; IIA would make the regulation counterfactuals mechanical) and developmental (this is the empirical-IO core you want to own). [Coven \[2025\]](#) and [Barbieri and Dobbels \[2026\]](#) both estimate full RC housing demand, so there is a direct, recent template.

2. Unit of demand — product $\neq$ jurisdiction. The single most important conceptual move: separate the *product* (what a household chooses) from the *jurisdiction* (a characteristic of that product). Households choose a **sub-jurisdictional location** — a Census tract, block group, or a constructed “neighborhood” — and the regulating **jurisdiction** enters as an attribute of that location, carrying the policy primitives $(\bar{z}_j, \tau_j)$. The mismatch your paper is about — households choose a *place*, shaped by a local government they did not choose at the parcel level — is then *built into the geography of the model* rather than assumed away. This is precisely the Bayer–Ferreira–McMillan design ([Bayer et al., 2007](#)), where products are houses/neighborhoods and jurisdiction boundaries enter through a boundary-discontinuity that also helps identification (§6). The market is a metro $\times$ year (or submarket $\times$ year); the choice set is the locations within it; the outside option is locating outside the metro (or a numeraire region).

3. Tenure choice — modeled, and not optional for you. Because the homevoter mechanism is an *owner* mechanism (the asset-value motive in the toy model is the capitalization of owners’ WTP), tenure cannot be conditioned away. The household’s choice is the *joint* discrete choice over (location $\times$ {own, rent-single-family, rent-multifamily}), exactly as in [Coven \[2025\]](#). This has three consequences developed in §3: the price object differs by tenure (sale price / user cost for owners; rent for renters); the random coefficients must include a tenure-specific taste; and the relevant “own-price” for the homevoter counterfactual is the owner-side capitalization, which the tenure split delivers directly.

2 The model

2.1 Choice problem

Households i in market t (metro/submarket $\times$ year) choose a location ℓ (tract/ neighborhood) and tenure $m \in \{o, r_s, r_m\}$ (own, rent single-family, rent multifamily). Indirect utility:

$$u_{i\ell mt} = \delta_{\ell mt} + \mu_{i\ell mt} + \varepsilon_{i\ell mt}, \quad (1)$$

$$\delta_{\ell mt} = x_{\ell t}\beta + \beta_m^{\text{ten}} - \bar{\alpha} P_{\ell mt} + \xi_{\ell mt}, \quad \mu_{i\ell mt} = \sum_k \sigma_k \nu_{ik} x_{\ell t}^k - \alpha_i^{\text{ten}} P_{\ell mt} + \Pi(d_i \times x_{\ell t}) - \kappa \text{movecost}_{i\ell}, \quad (2)$$

where $x_{\ell t}$ are location attributes (job access, school quality, density/“character”, crime, transit, structure mix), $P_{\ell mt}$ is the tenure-specific price (sale price or user cost for o ; rent for r_s, r_m), $\xi_{\ell mt}$ is unobserved location-tenure quality, d_i are demographics (income, family structure, age), $\nu_{ik} \sim N(0, 1)$ are idiosyncratic tastes, and ε is Type-I EV. The price coefficient is income-dependent, $\alpha_i^{\text{ten}} = \bar{\alpha} + \Pi_\alpha \text{income}_i$, which is what makes high-income households the natural demanders of low-density locations — the empirical analogue of the toy model’s θ -sorting. Moving costs $\kappa \text{movecost}_{i\ell}$

(estimated off bilateral migration data, as in [Coven, 2025](#), who uses Infutor/Verisk migration flows) discipline the geographic substitution pattern; without them, cross-location elasticities are unrealistic.

2.2 Estimation

Standard BLP/pyblp pipeline ([Conlon and Gortmaker, 2020](#), [Nevo, 2000](#)):

1. Berry inversion ([Berry, 1994](#)) maps observed location-tenure shares to mean utilities $\delta_{\ell mt}$.
2. GMM on the moment $E[\xi_{\ell mt} | Z] = 0$, with instruments Z for the endogenous price (§5).
3. Micro-moments ([Petrin, 2002](#)) matching demographic-attribute covariances (e.g. income $\times$ density, family size $\times$ tenure) from the ACS PUMS, to pin down the Π, σ heterogeneity that the macro shares alone identify weakly.
4. Optimal instruments / differentiation IVs ([Gandhi and Houde, 2019](#)) to sharpen the random coefficients.

3 Tenure choice in detail

Three modeling decisions and the literature on each:

- **Nesting vs. joint choice.** A nested structure (location at the top, tenure within) imposes a correlation pattern; a fully joint choice with a tenure random coefficient is more flexible and is what [Coven \[2025\]](#) adopts. Recommend the joint choice with a tenure-specific random coefficient, falling back to a nest only if identification is thin.
- **The owner price object.** Owners do not pay “price”; they pay *user cost* ([Poterba, 1984](#)): price $\times$ (after-tax interest + property tax + maintenance + depreciation – expected appreciation). You must build a user-cost series, not use raw sale prices, or the own/rent margin is mis-measured. This is a real data task (§4).
- **Why tenure interacts with the homevoter story.** The counterfactuals perturb regulation and ask how owner asset values move. Only the owner branch of demand capitalizes regulation into prices; the renter branch capitalizes it into rents (incidence differs). The tenure split is therefore not a refinement — it is the channel through which Layer I feeds the homevoter objective in Layer II. Tenure choice also responds endogenously to institutional-investor presence and to regulation, which is exactly the margin [Barbieri and Dobbels \[2026\]](#) and [Coven \[2025\]](#) exploit; if you ever want the build-to-rent angle, the architecture already supports it.

4 Data sources

Object	Source	Notes / template
Prices & rents (by tenure), transactions, structure attributes	CoreLogic tax & deed (you have this); ACS for rent distributions; optionally Zillow ZTRAX/listings	Barbieri–Dobbels scrape Zillow for house-level rents/prices; CoreLogic is your spine
Household demographics, tenure shares, income groups	ACS PUMS (micro) + ACS tract tables (shares)	Coven estimates demand by income group from ACS
Migration flows (moving costs)	Infutor / Verisk address-history; IRS county-to-county as backup	Coven uses bilateral migration to estimate moving costs
Job access / labor-market access	LEHD LODES (jobs by block); commute matrices	Feeds the $w_m = w(N)$ agglomeration coupling (Layer III)
Amenities (schools, crime, transit, open space)	CA Dept. of Education; transit GTFS; parks/open-space layers	Open space is an <i>amenity control</i> , not an instrument
Soil / geology (the instrument)	USDA SSURGO (USDA Natural Resources Conservation Service, 2024); CGS liquefaction & seismic zones (California Geological Survey, 2024); USGS	See §5
Construction-cost schedule	RS Means residential cost data; foundation-type cost differentials	Maps soil class $\rightarrow$ \$/unit foundation premium
Regulation $(\bar{z}_j, \tau_j)$	Your minutes dataset + LLM zoning envelope	The project’s own contribution (Layer II)

The CoreLogic spine is the same data class both template JMPs lean on ([Coven, 2025](#) uses Verisk property files; [Gorback et al., 2024](#) and [Barbieri and Dobbels \[2026\]](#) use CoreLogic/property-tax records), so you are on a well-trodden data path for everything except the soil instrument and the regulation measure.

5 The soil cost-shifter instrument

5.1 What it must do, and the precedent

The instrument must shift the tenure-specific price $P_{\ell mt}$ (through construction/supply cost) while being excludable from $\xi_{\ell mt}$ (location-tenure desirability). The lineage is geology-as-cost-shifter: [Rosenthal and Strange \[2008\]](#) instrument employment density with geological features — access to bedrock, seismic and landslide hazard, water coverage — to identify agglomeration; [Combes et al. \[2010\]](#) use geology and history similarly; [Saiz \[2010\]](#) uses terrain (slope, water) for MSA-level supply elasticity; [Barr et al. \[2011\]](#) use depth-to-bedrock as a construction-cost shifter for building height. **So the ingredients are not novel**; the novelty is the *application* (instrumenting price in a tenure-differentiated sorting model) and the *residualization* that isolates the cost-only component.

Remark 1 (The exclusion threat is your own model). [Rosenthal and Strange \[2008\]](#)’s *entire finding is that geology shifts* agglomeration. *Your toy model has agglomeration ($w_m = w(N)$) entering*

location value. So raw geology is correlated with ξ through the agglomeration channel — the same channel a whole literature exploits. You cannot cite Rosenthal–Strange to validate the cost channel and simultaneously ignore that their result is an exclusion violation for you. The instrument must be the soil component orthogonal to agglomeration-relevant geography (seismic risk, water access, terrain, distance to job centers).

5.2 Construction, step by step

1. **Raw soil engineering properties.** From SSURGO map units, pull the variables that drive *foundation cost*: shrink-swell potential (linear extensibility / COLE), plasticity index, bearing capacity / load class, depth to restrictive layer, drainage class. These are the expansive-clay / bearing-capacity / shrink-swell dimensions. The Bay Area variation is large and salient (bay mud and fill vs. expansive East Bay clays vs. competent alluvium/bedrock).
2. **Map to a cost premium.** Translate soil class into a predicted per-unit foundation/ construction premium using an external engineering cost schedule (RS Means foundation-type differentials; geotech rules of thumb for pile/pier/mat vs. slab-on-grade). This yields a *continuous predicted-cost index* $\hat{c}_\ell$, not a categorical soil dummy — better first-stage behavior and a cleaner economic interpretation. This is the [Barr et al. \[2011\]](#) move (geology $\rightarrow$ predicted cost), adapted to soil.
3. **Residualize against the amenity & agglomeration channels (the crux).** Regress $\hat{c}_\ell$ on the confounders that load into ξ — distance to shore/water, slope/ elevation, open-space proximity, seismic/liquefaction hazard, distance to job centers — and use the *residual* (or include those confounders as controls in the second stage and the instrument as the orthogonalized part). What remains is soil-driven cost variation plausibly unrelated to taste. This is the step that converts a known-and-contaminated instrument into a defensible one.
4. **Aggregate to the location unit and (optionally) interact with a time shifter.** Average to tract/neighborhood; optionally interact with a time-varying construction-cost or interest-rate series to generate within-location variation (a Bartik-flavored move, [Bartik, 1991](#), [Goldsmith-Pinkham et al., 2020](#)), since a pure cross-sectional soil instrument identifies only cross-sectional price variation.

5.3 Threats and checks

- **Relevance.** Soil moves cost a few percent, not orders of magnitude ([Barr et al., 2011](#) found bedrock’s skyline effect *small*). Report the first-stage F ; if weak, lean on the time-interacted version and consider many-instrument-robust inference.
- **Seismic capitalization.** Liquefaction zones may be priced as risk (a ξ channel), so seismic hazard must be a *control*, never part of the instrument. Argue (and show) the residual cost variation is orthogonal to seismic risk.
- **Soil $\leftrightarrow$ amenity collinearity.** Bay mud *is* waterfront; fill *is* flat-and-central. Hence the residualization in step 3 is not optional.
- **Soil $\leftrightarrow$ regulation (the back-door double-spend).** If a jurisdiction has a geologic-hazard combining district that regulates *on soil*, then soil moves τ_j and the instrument is contaminated by the very outcome of Layer II. Check for soil-based overlays before committing; prefer the foundation-cost dimension least likely to carry its own overlay.
- **Over-identification.** With the BLP differentiation IVs ([Gandhi and Houde, 2019](#)) *plus* the soil instrument, you can run an overid test and report instrument-by-instrument first stages — useful for the referee and for separating the soil instrument’s identifying content.

6 Identification

The price endogeneity is $\text{Cov}(P_{\ell mt}, \xi_{\ell mt}) \neq 0$. Three identifying strategies, used together (Berry and Haile, 2014 on what market-level data can identify):

1. **Soil cost-shifter** (§5) — your contribution, the supply-cost IV.
2. **Boundary discontinuities** (Bayer et al., 2007) — comparing locations across a jurisdiction boundary holds fixed the slowly-varying ξ while the regulatory regime jumps; natural in your fragmented-jurisdiction setting and a second identification source.
3. **Differentiation / BLP instruments** (Gandhi and Houde, 2019, Berry et al., 1995) — attributes of competing locations, for the random coefficients specifically.

Remark 2 (Do not spend the preemption ramp here). *The preemption ramp is reserved for the regulation game (Layer II). Using it to instrument demand prices and to identify the strategic slope double-spends one exclusion restriction. The soil instrument exists precisely so demand has its own supply-cost shifter, independent of preemption. Keep them separate.*

7 Key judgment calls

Spatial unit & choice-set size

Tract vs. block group vs. constructed neighborhood. Finer = better match to true choice but explodes the choice set and thins shares (small-share bias; Gandhi and Houde, 2019 matters more). Coarser = stable shares but blurs the jurisdiction mismatch you care about. *Likely resolution:* block group or a school-zone-based neighborhood, with sampled choice sets if needed.

User cost vs. price for owners

Build a proper user-cost series (Poterba, 1984, Sinai and Souleles, 2005) or proxy with price? Using raw price mis-states the own/rent margin and contaminates tenure substitution. *Recommend* building user cost; it is a few weeks of work and load-bearing.

Market definition / outside option

What is “the market” (metro-year? submarket-year?) and the outside option (live outside the Bay Area? a numeraire region?). This sets the level of substitution the price elasticity captures and must align with the submarket structure of the toy model.

Heterogeneity dimensions

Which demographics get random coefficients / interactions: income (essential, for θ -sorting), family structure, age, race (Bayer-style)? More heterogeneity = richer counterfactuals but heavier micro-moment demands on the ACS.

Static vs. dynamic demand

A static RC model ignores forward-looking moving/tenure decisions. Both template JMPs are essentially static with moving costs. *Recommend* static-with-moving-costs for the headline; dynamic is an extension, consistent with the toy model’s own scoping.

Endogenous amenities

If amenities respond to who sorts in (Diamond, 2016, Almagro and Domínguez-Iino, 2024), ξ has an endogenous-amenity component. Decide whether to model this or treat amenities as exogenous-conditional-on-controls for the first pass. *Recommend* exogenous for v1, flagged as extension.

8 What to walk into the meeting with

1. **Confirm the product unit** (block group / school-zone neighborhood) and that jurisdiction enters as an attribute — the move that encodes your paper’s central mismatch.
2. **Present the soil instrument honestly**: ingredients from Rosenthal–Strange lineage, novelty in the application and the residualization, with the agglomeration-channel threat (Remark 1) stated up front. Guren will respect the pre-emption of his own objection.
3. **Get a ruling on user cost** — whether he thinks the owner-side user-cost build is worth it or whether a simpler proxy suffices for the counterfactuals.
4. **Confirm the instrument division of labor**: soil for demand prices, preemption for the regulation game; never the same shifter twice.
5. **Show the two template JMPs** (Coven, 2025, Barbieri and Dobbels, 2026) and ask which estimation choices to imitate vs. improve — both are close enough that “what would you do differently than Coven” is a productive, concrete question.

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